ORIGINAL ARTICLE

Clinical outcomes of endoscopic papillectomy for ampullary adenoma and adenocarcinoma: a multicenter retrospective study in Japan

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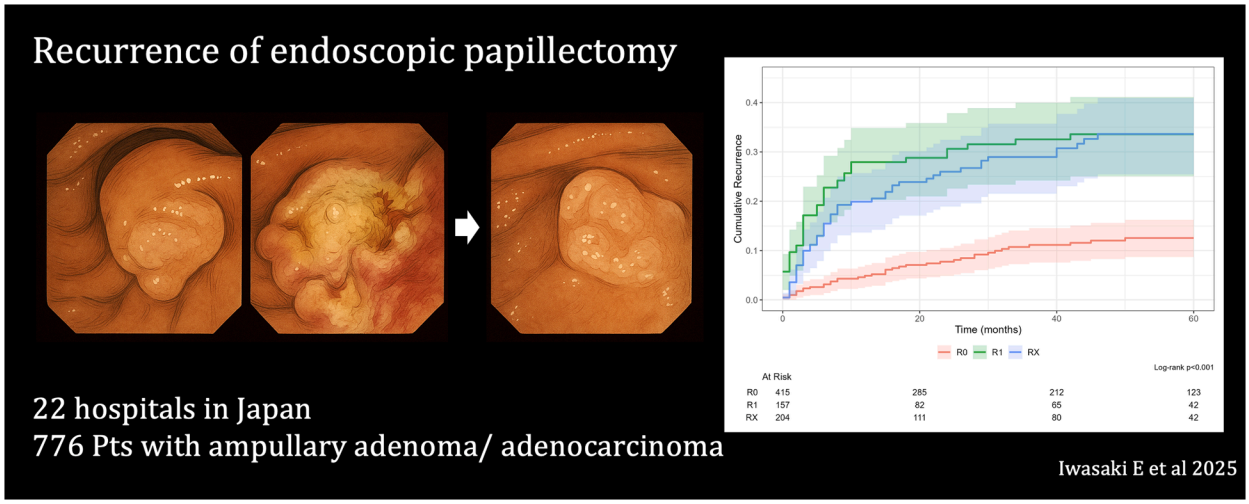
GRAPHICAL ABSTRACT



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Background and Aims: Endoscopic papillectomy (EP) is a minimally invasive treatment for ampullary epithelial tumors. However, clinical outcomes and predictors of recurrence remain unclear.

Methods: We conducted a multicenter, retrospective cohort study at 22 tertiary centers in Japan, including patients who underwent EP between January 2009 and December 2020. The primary outcome was cumulative recurrence in patients with histologically confirmed adenoma or adenocarcinoma. Recurrence predictors were evaluated using Cox proportional hazards model.

Results: Of 875 patients, 798 had confirmed ampullary epithelial tumor, and 776 were included in the long-term outcome analysis. The 3-year cumulative recurrence rate was 20%. Indeterminate (Rx) and positive (R1) resection margins were independently associated with higher recurrence compared with negative margins (R0), with hazard ratios of 2.30 (95% CI, 1.44-3.68) and 2.95 (95% CI, 1.85-4.71), respectively. R0 resection was achieved in 52.1% (416/798) of patients, with a 3-year cumulative recurrence rate of 11% (95% CI, 7.6%-15%).

Conclusions: Recurrence after EP for ampullary epithelial tumors is common, even after R0 resection. These findings highlight the need for long-term surveillance following EP. (Gastrointest Endosc 2025;■:1-12.)

INTRODUCTION

Ampullary tumors are rare and predominantly epithelial in origin, primarily consisting of adenomas and adenocarcinomas. Increased use of routine upper endoscopies and greater awareness among endoscopists have led to greater detection rates of papillary neoplasms.[1] Historically, pancreaticoduodenectomy and transduodenal surgical ampullectomy were the preferred treatments,[2] but endoscopic papillectomy (EP) has recently become the favored approach as a result of its minimally invasive nature.[3]

Guidelines recommend EP as the first-line therapy for ampullary adenomas without intraductal invasion.[4-6] However, the limited diagnostic accuracy of pretreatment duodenal biopsy[7,8] often leads to EP being performed for lesions later identified as adenocarcinomas.[9,10] Nevertheless, the long-term outcomes of EP, including recurrence rates and predictors, remain poorly defined because of a lack of large-scale studies. To address this gap, we conducted a nationwide, multicenter retrospective study in collaboration with the Japanese Endoscopy Society's EP guideline committee and multiple high-volume centers to evaluate long-term recurrence rates and identify predictors of recurrence after EP.

METHODS

Study design and setting

This retrospective, multicenter observational cohort study was conducted at 22 tertiary referral centers across Japan. All patients who underwent EP between January 1, 2009, and December 31, 2020, were included. The study was conducted under an opt-out policy approved by the institutional review board at each participating center and was registered in the Clinical Trials Registry (UMIN 000040288). Data were collected from each center using a centralized web-based registry. This study was approved by the ethics committee of Keio University Hospital.

Study patients

Among patients who underwent EP, those with nonepithelial ampullary tumors on the basis of postprocedural pathology were excluded. In addition, patients who underwent surgery after EP because of a confirmed or suspected residual adenocarcinoma, as well as those who experienced EP-related mortality, were excluded from the long-term recurrence analysis. Written informed consent for EP was obtained from all patients, and the study used an opt-out design for data collection.

Data collection

The collected data included patient demographics (age and sex), Charlson Comorbidity Index, history of familial adenomatous polyposis (FAP), use of antithrombotic agents, and use of use of acid-suppressive medications (proton-pump inhibitors and H_2 receptor antagonists). Tumor characteristics were recorded, including tumor size as measured endoscopically, presence of laterally spreading tumor involving the papilla (defined as $\geq$10 mm extension), and the diameter and intraductal extension of bile and pancreatic duct as assessed by cross-sectional imaging or endoscopic ultrasound. Preoperative diagnoses were determined on the basis of endoscopic biopsy results. Moderate-to-severe comorbidity was defined as a Charlson Comorbidity Index score >3.[11] Bile duct and pancreatic duct dilatation were defined as diameters >8 mm and >4 mm, respectively. The EP treatment eras were divided into 2 time frames on the basis of the median period: 2009 to 2014 (first half) and 2015 to 2020 (second half). In addition, variables related to the EP procedure, outcomes, and adverse events were collected, as detailed herein.

EP procedure

The EP was performed during inpatient admission by experienced teams at all participating centers via the approach most commonly used in Japan.[12] A duodenoscope was used to snare the lesion, and submucosal injection of a lifting agent was performed in patients with a laterally spreading component. After snaring, polypectomy was performed using either a mixed current or a pure cutting electrocautery mode. Prophylactic drainage of the bile duct involved placement of a biliary stent with or without sphincterotomy, or sphincterotomy alone. For pancreatic duct, similarly, prophylactic pancreatic tube placement with/without pancreatic sphincterotomy or only pancreatic sphincterotomy without tube placement was used. If there was residual tumor detected endoscopically, additional argon plasma coagulation (APC) was carried out. In order to prevent bleeding, mucosal defect closure with clips was also performed. Detailed procedural data were collected from each center, including stent type, shape, size, and length, duration of stent placement, and extent of clip closure (none, partial <50%, or sufficient ≥50%). Resection methods were categorized as en bloc resection (entire lesion removed in a single piece) and piecemeal resection (lesion removed in multiple pieces). Technical success was defined endoscopically as complete tumor removal with no visible residual lesion at the time of index procedure. Experienced-operator procedure was defined as EP performed by an endoscopist with ≥20 previous cases. Pathologic resection status was classified as follows: R0—complete resection by en bloc with negative horizontal and vertical margins; Rx—indeterminate margins including both unclear margin regardless of whether it is a complete resection by en bloc or piecemeal resection and negative margin by piecemeal resection; and R1—positive margins.[13] Pathologic assessment also included evaluation of bile or pancreatic duct involvement (categorized as positive or nonpositive, the latter including both negative and undetermined cases), and the final pathologic diagnosis (defined as adenoma, carcinoma in adenoma or adenocarcinoma).

Specimens were typically fixed and serially sectioned at 2-mm intervals, followed by histologic evaluation with hematoxylin–eosin staining. If the bile or pancreatic duct margin was not adequately represented, additional deeper sections were prepared. All histologic assessments were conducted by pathologists at each participating center and collected as follows: adenoma unknown, adenoma mild atypia, adenoma moderate atypia, adenoma severe atypia, carcinoma in adenoma, adenocarcinoma (pathologists in the West consider noninvasive intraepithelial neoplasia to be a low- or high-grade adenoma/dysplasia, as it is a lesion that has no possibility of metastasis, whereas in Japan, pathologists diagnose it as adenoma or carcinoma, regardless of the presence or absence of invasion).[14]

Clinical indications for EP

The primary indication for EP was adenoma without intraductal extension. For adenomas, follow-up without additional surgery was generally considered acceptable, regardless of whether the resection was R0, Rx, or R1. By contrast, for adenocarcinoma or carcinoma in adenoma, additional surgery was recommended when the resection margin was Rx or R1. Even after R0 resection, Tis–T1a[15] lesions confined to the ampulla of Vater without involvement of the sphincter of Oddi were monitored, whereas T1a lesions involving the sphincter of Oddi, as well as T1b or deeper lesions, were referred for surgery. In all cases, the final treatment strategy was determined through thorough discussion with the patient.

Outcomes

The primary outcome was cumulative recurrence during follow-up among patients pathologically diagnosed with adenoma or adenocarcinoma. Recurrence was defined as histologic confirmation of neoplastic tissue. Although follow-up protocols varied by center and clinical context, endoscopic surveillance was typically performed every 6 to 12 months. Factors associated with recurrence in the overall cohort and in the R0 resection subgroup were analyzed.

Secondary outcomes included length of hospitalization, adverse events, and retreatment for local recurrence, with clinical success after initial recurrence defined as adenoma or adenocarcinoma not being detected by biopsy during the follow-up period. Adverse events included pancreatitis, immediate or delayed bleeding, perforation, cholangitis, late-onset strictures, and procedure-related mortality. Definitions and severity followed the American Society for Gastrointestinal Endoscopy classification.[16]

Statistical analysis

Continuous variables were summarized as medians with IQR and compared using the Mann–Whitney *U* test. Categorical variables were summarized as percentages and compared using the χ^2 test or Fisher exact test, as appropriate. Cumulative recurrence rates were assessed with Kaplan-Meier curves and compared using the log-rank test. Patients were censored at the time of 5 years post-EP if recurrence had not occurred by that time in accordance with established methodology.[17] Cox proportional hazards models were used to evaluate covariate having impact on recurrence including age (≥65 years), sex, FAP, larger tumor size (≥15 mm), laterally spreading tumor involving the papilla, resection type (en bloc or piecemeal), operator experience (≥20 cases were defined as expert), treatment eras, submucosal injection, use of APC for residual lesions, pathologic resection status (R0/R1/Rx), post-EP pathologic involvement of the bile duct or pancreatic duct, and the final pathologic

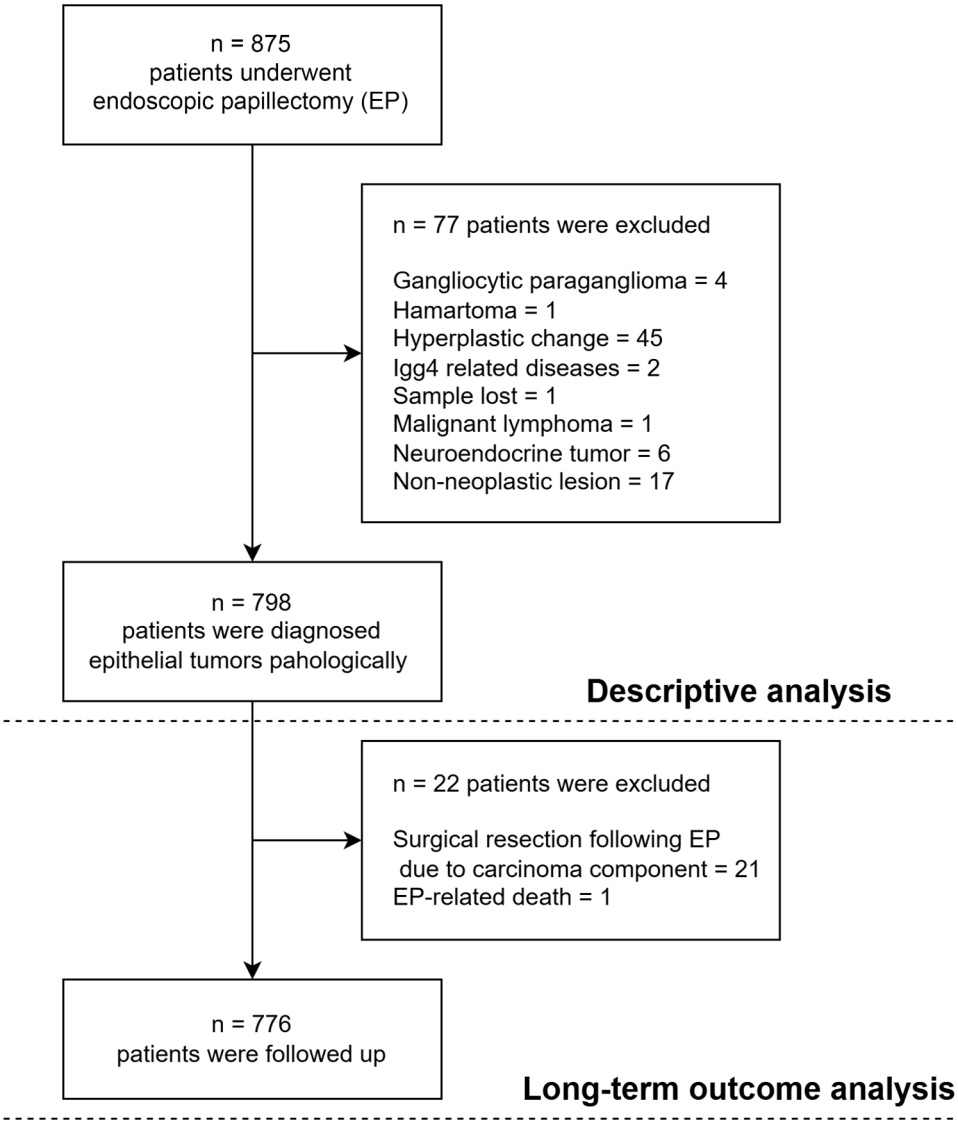

Figure 1. Flowchart of patient inclusion.

diagnosis (defined as adenoma, carcinoma in adenoma or adenocarcinoma). These covariates were chosen on the basis of clinical perspective and previous studies and cut-off values for age and tumor size were on the basis of their median values. For the subgroup analysis, we calculated the cumulative recurrence rates for each final pathologic diagnosis group (adenoma, carcinoma-in-adenoma, and adenocarcinoma). For recurrence analysis restricted to R0 resections, the same variables were included. A P value <.05 was considered statistically significant. All analyses were performed with R, version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Patients and procedure characteristics

A total of 875 cases were enrolled; 798 were confirmed pathologically as adenoma or adenocarcinoma after EP. Seventy-seven cases were excluded (Fig. 1). Baseline clinical characteristics are summarized in Table 1. Of the 798 cases, 53 patients (6.6%) had genetically confirmed FAP.

Among 798 patients with confirmed adenoma or adenocarcinoma, en bloc resection was achieved in 710 (89.0%), whereas 88 patients (11.0%) underwent piecemeal resection. Technical success—defined as complete removal of the lesion at the time of endoscopy—was achieved in 94.9%.

TABLE 1. Patients and procedure characteristics

	Missing	*N* = 798
Age, y	0 (0%)	67.0 (58.0, 73.0)
Female	0 (0%)	320 (40.1%)
Familial adenomatous polyposis	0 (0%)	53 (6.6%)
CCI score	0 (0%)	0 (0, 1)
Antithrombotic agent	0 (0%)	4 (0.5%)
Agents for peptic ulcer	1 (0.1%)	
Proton pomp inhibitors		458 (57.5%)
Histamine H2-receptor antagonists		16 (2.0%)
None		323 (40.5%)
Tumor size, mm	0 (0%)	15 (10-20)
LST-p	0 (0%)	121 (15.2%)
Diameter of bile duct, mm	115 (14%)	6.5 (5.0, 8.0)
Preoperative bile duct invasion	0 (0.0%)	
Positive		30 (3.8%)
Nonpositive		768 (96.2%)
Diameter of pancreatic duct, mm	115 (14%)	2.0 (2.0, 3.0)
Preoperative pancreatic duct invasion	0 (0.0%)	
Positive		10 (1.3%)
Nonpositive		788 (98.7%)
Preoperative histology	7 (0.9%)	
Adenocarcinoma		65 (8.2%)
Carcinoma in adenoma		13 (1.6%)
Adenoma severe atypia		90 (11.4%)
Adenoma moderate atypia		167 (21.1%)
Adenoma mild atypia		58 (7.3%)
Adenoma unknown		395 (49.9%)
Others		3 (0.4%)
Endoscopically technical success	0 (0%)	757 (94.9%)
Resection methods	0 (0%)	
En bloc resection		710 (89.0%)
Piecemeal resection		88 (11.0%)
Pathologic resection status	0 (0%)	
R0 resection		416 (52.1%)
R1 resection		176 (22.1%)
Rx resection		206 (25.8%)
Pathologic vertical margin	0 (0%)	
Negative		528 (66.2%)
Positive		111 (13.9%)
Undetermined		159 (19.9%)
Pathologic horizontal margin	0 (0%)	
Negative		543 (68.0%)
Positive		96 (12.0%)
Undetermined		159 (19.9%)
Pathologic bile duct invasion	0 (0%)	
Positive		29 (3.6%)
Nonpositive		769 (96.4%)

(*continued on the next page*)

TABLE 1. Continued

	Missing	*N* = 798
Pathologic pancreatic duct invasion	0 (0%)	
Positive		9 (1.1%)
Nonpositive		789 (98.9%)
Final pathologic diagnosis	0 (0%)	
Adenoma		618 (77.4%)
Carcinoma in adenoma		88 (11.0%)
Adenocarcinoma		92 (11.5%)
Submucosal injection	0 (0%)	146 (18.3%)
Expert procedure	0 (0%)	509 (63.8%)
Procedure date	0 (0%)	362 (45.3%)
First period: 2009-2014		362 (45.3%)
Second period: 2015-2020		436 (54.6%)
Cut setting for high-frequency device	1 (0.1%)	
Mixed current mode		442 (55.5%)
Pure cutting mode		355 (44.5%)
Additional coagulation using APC	0 (0%)	25 (3.1%)
Prophylactic pancreatic duct tube	0 (0%)	747 (93.6%)
Pancreatic sphincterotomy	0 (0%)	63 (7.9%)
Prophylactic bile duct tube	0 (0%)	658 (82.5%)
Biliary sphincterotomy	3 (0.4%)	127 (16.0%)
Prophylactic clipping closure	0 (0%)	
Partial closure (<50%)		395 (49.5%)
Sufficient closure (≥50%)		182 (22.8%)

Values are median (range, IQR) or n (%).
APC, Argon plasma coagulation; *CCI*, Charlson Comorbidity Index; *LST-p*, laterally spreading tumor involving the papilla.

R0 resection was achieved in 416 patients (52.1%). Among en bloc resections, only 58.6% (416/710) were ultimately classified as R0, with the remainder designated Rx or R1 due to thermal artifact or margin involvement.

Preresection evaluation for intraductal extension was performed using EUS in 81.5% (650/798) of cases. Among the remaining cases, 132 were evaluated with other imaging modalities including ERCP, computed tomography, or magnetic resonance imaging, whereas 18 patients underwent EP without dedicated evaluation. Among patients who underwent these preresection evaluations, bile duct extension in 3.8% (30/780) and pancreatic duct extension in 1.3% (10/780) were confirmed. Pathologic examination confirmed bile duct extension in 3.6% (29/798) and pancreatic duct extension in 1.1% (9/798). The R0 resection rate for patients with positive bile duct or pancreatic duct extension was 20.4% (28/137).

Additional details related to procedure are provided in Supplementary Table 1, available online at www.giejournal.org. There were procedure differences including pathologic resection status, expert procedure, and additional coagulation using APC between 2 periods and are provided in Supplementary Table 2, available online at www.giejournal.org. A summary of recurrence events stratified by covariates is presented in Supplementary Table 3, available online at www.giejournal.org.

Primary outcome

Of the 798 patients with confirmed adenoma or adenocarcinoma, 776 were included in the primary analysis. Twenty-one patients were excluded due to undergoing surgery for confirmed or suspected residual adenocarcinoma, and 1 was excluded because of procedure-related mortality (Fig. 1, Supplementary Fig. 1, available online at www.giejournal.org). The median follow-up duration was 33 months (IQR, 6.75-60 months).

Tumor recurrence occurred in 18.3% (142/776) of patients, with a median time to recurrence of 6 months (IQR 2-18). The vast majority were local recurrences (139/142; 97.9%), whereas 3 cases (2.1%) were metastatic (1 R1 resection and 2 R0 resection cases).

The cumulative recurrence rate was 13% (95% CI, 11%-16%) at 1 year, 20% (95% CI, 17%-23%) at 3 years, and 22% (95% CI, 19%-26%) at 5 years (Fig. 2). Multivariable Cox

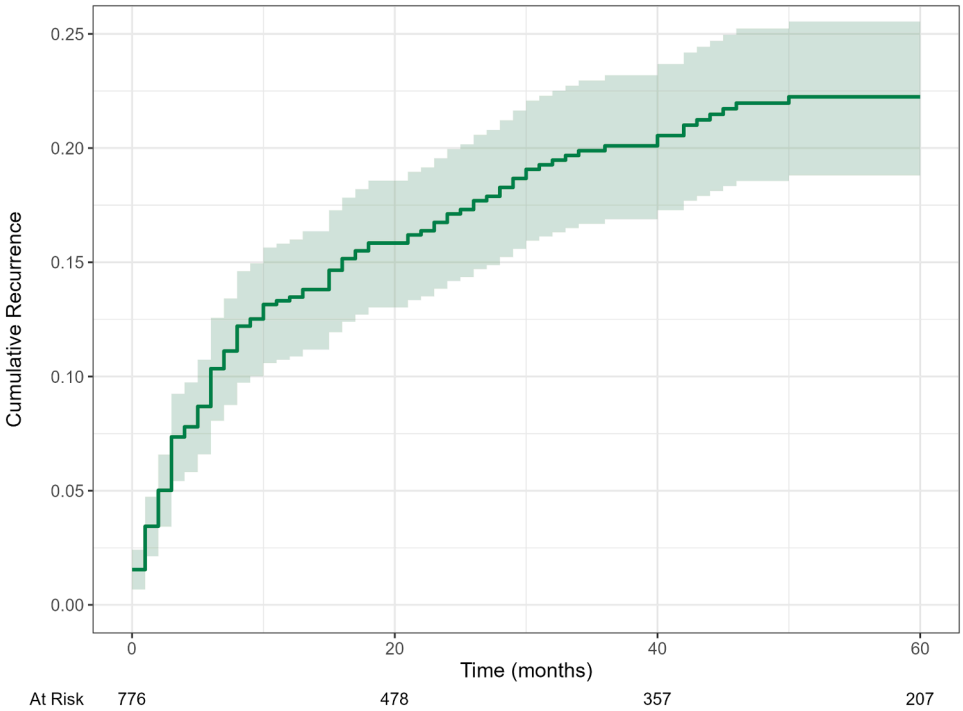

Figure 2. Kaplan-Meier curve of cumulative recurrence after endoscopic papillectomy for ampullary epithelial tumors.

TABLE 2. Cox-proportional hazards model for recurrence

	Univariate			Multivariate		
Characteristic	**HR**	**95% CI**	***P* value**	**HR**	**95% CI**	***P* value**
Elderly (age ≥ 65 y)	1.19	0.85-1.67	.3	1.31	0.90-1.90	.2
Female	1.52	1.09-2.12	**.015**	1.48	1.06-2.08	**.022**
FAP	1.92	1.14-3.23	**.014**	1.59	0.90-2.83	.11
Large tumor (≥15 mm)	1.27	0.91-1.77	.2	0.88	0.60-1.28	.5
LST-p	1.86	1.26-2.74	**.002**	1.39	0.91-2.12	.13
Piecemeal resection	3.04	2.08-4.44	**<.001**	1.69	1.07-2.65	**.023**
Expert procedure	1.05	0.74-1.50	.8	1.15	0.79-1.67	.5
Procedure period (2015-2020)	1.56	1.10-2.21	**.012**	1.57	1.09-2.26	**.016**
Submucosal injection	1.19	0.79-1.79	.4	0.97	0.64-1.48	.9
Argon plasma coagulation	1.78	0.87-3.63	.11	1.54	0.73-3.29	.3
Pathologic resection status						
Margin negative (R0)						
Margin positive (R1)	3.71	2.43-5.68	**<.001**	2.95	1.87-4.66	**<.001**
Margin undetermined (Rx)	3.21	2.12-4.86	**<.001**	2.38	1.49-3.81	**<0001**
Pathologic invasion to bile duct	2.03	1.40-2.94	**<.001**	1.46	0.61-3.50	.4
Pathologic invasion to pancreatic duct	1.95	1.32-2.87	**<.001**	2.58	0.81-8.27	.11
Final pathologic diagnosis						
Adenoma						
Carcinoma in adenoma	1.05	0.60-1.84	.9	1.07	0.60-1.91	.8
Adenocarcinoma	2.00	1.28-3.12	**.002**	1.84	1.16-2.92	**.009**

Bold text indicates statistical significance ($P < .05$).
FAP, Familial adenomatous polyposis; *HR*, hazard ratio; *LST-p*, laterally spreading tumor involving the papilla.

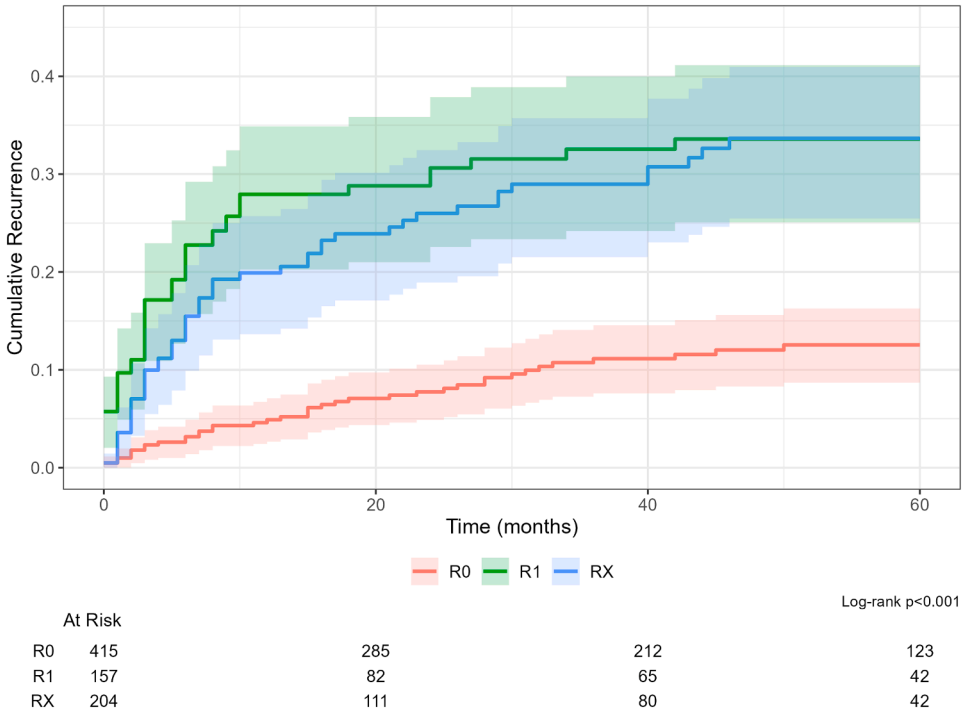

Figure 3. Kaplan-Meier curves of cumulative recurrence after endoscopic papillectomy for ampullary epithelial tumors stratified by pathologic resection status (R0, R1, Rx).

regression identified the following as independent predictors of recurrence (Table 2): female sex, piecemeal resection, R1 or Rx pathologic margin status, adenocarcinoma, and EP performed in the second half of the study period (2015-2020).

Among patients with adenoma, carcinoma-in-adenoma, and adenocarcinoma, the 3-year cumulative recurrence rates were 19% (95% CI, 15%-22%), 20% (95% CI, 16%-24%), and 25% (95% CI, 20%-30%), respectively. Kaplan–Meier curves (Supplementary Fig. 2, available online at www.giejournal.org) showed significant differences among the 3 groups (log-rank $P < .001$).

Pathologic margin status was the strongest predictor of recurrence. Log-rank analysis showed significant differences in recurrence rates by margin category ($P < .001$; Fig. 3):

- R0 resection: 4.9% at 1 year, 11% at 3 years, and 13% at 5 years;
- R1 resection: 28% at 1 year, 33% at 3 years, and 34% at 5 years; and
- Rx resection: 20% at 1 year, 29% at 3 years, and 34% at 5 years.

The recurrence rates within 6 months were 3% (12/415) for R0 resections, 13.2% (27/204) for Rx resections, and 21.7% (34/157) for R1 resections. The median time to recurrence was 15 months for R0, 6 months for Rx, and 3 months for R1 resections.

In the subgroup of patients with R0 resection, female sex, use of APC, pathologic biliary invasion and adenocarcinoma were significantly associated with increased recurrence risk (Table 3). Kaplan–Meier curves demonstrated significant differences among the 3 groups (adenomas, carcinoma in adenoma, and adenocarcinoma) (log-rank $P < .001$, Supplementary Fig. 3, available online at www.giejournal.org).

Secondary outcome

The median length of hospitalization was 13.0 days (IQR 10.0-16.0). Adverse events occurred in 32.5% (259/798) of patients, with the most common delayed bleeding (19.4%) and pancreatitis (13.8%) (Table 4). Severe adverse events occurred in 3.9% of patients. One treatment-related death (0.1%) was reported in a patient with cirrhosis who died from liver failure after delayed postprocedural bleeding.

Among patients with local recurrence, retreatment modalities included endoscopic resection (39/139; 31.0%), APC (56/139; 44.4%), surgery (14/139; 11.1%), and trimming with endoscopic forceps (2/139; 1.6%). Data were missing for 13 patients (9.4%). Endoscopic retreatment achieved clinical success in 74.5% (70/97) of cases.

DISCUSSION

The novelty of our study stems from its large, multicenter design and the detailed evaluation of recurrence risk according to resection status (R0, R1, and Rx). Our

TABLE 3. Cox-proportional hazards model for recurrence in R0 resection group

	Univariate			Multivariate		
Characteristic	**HR**	**95% CI**	***P* value**	**HR**	**95% CI**	***P* value**
Elderly (age ≥ 65 y)	1.21	0.64-2.29	.6	1.46	0.72-2.95	.3
Female	2.14	1.13-4.05	**.019**	2.09	1.07-4.07	**.032**
FAP	3.20	1.25-8.18	**.015**	2.90	0.95-8.89	.063
Large tumor (≥15 mm)	1.34	0.71-2.52	.4	1.22	0.61-2.45	.6
LST-p	2.38	1.05-5.40	**.037**	2.12	0.83-5.43	.12
Expert procedure	0.73	0.39-1.39	.3	0.87	0.43-1.76	.7
Procedure period (2015-2020)	1.54	0.81-2.93	.2	1.42	0.73-2.76	.3
Submucosal injection	1.00	0.42-2.40	>.9	0.79	0.31-2.02	.6
Argon plasma coagulation	4.08	0.98-17.0	.053	6.26	1.44-27.1	**.014**
Pathologic invasion to bile duct	2.10	0.82-5.37	.12	4.86	1.08-21.8	**.039**
Pathologic invasion to pancreatic duct	1.43	0.44-4.64	.6	0.00	0.00-Inf*	>.9
Final pathologic diagnosis						
Adenoma						
Carcinoma in adenoma	1.05	0.60-1.84	0.9	1.09	0.37-3.26	0.9
Adenocarcinoma	2.00	1.28-3.12	**.002**	2.74	1.22-6.14	**.015**

Bold text indicates statistical significance ($P < .05$).
FAP, Familial adenomatous polyposis; *HR*, hazard ratio; *LST-p*, laterally spreading tumor involving the papilla.
*Since the denominator is zero, the odds ratio is infinity.

findings show a recurrence rate of 18.3% (142/776) with a median time to recurrence of 6 months. The 3-year cumulative recurrence rate was 20% (95% CI, 17%-23%). R0 resection was achieved in 52.1% (416/798) of patients, with a 3-year cumulative recurrence rate of 11% (95% CI, 7.6%-15%). These findings indicate that recurrence is not uncommon even after R0 resection. Therefore, ongoing surveillance following EP is warranted, even in R0 cases.

Reported recurrence rates following EP in previous studies range from 0% to 32.7%,[9,12,18-20] consistent with our findings. In our cohort, R1 and Rx resection margins were the strongest predictors of recurrence when compared with R0 (hazard ratio for Rx: 2.30; 95% CI, 1.44-3.68; hazard ratio for R1: 2.95; 95% CI, 1.85-4.71). This is clinically intuitive, as negative histologic margins reduce the likelihood of residual disease. In this study, recurrence within 6 months was observed in 3% of R0, 13.2% of Rx, and 21.7% of R1 cases. Because Rx and R1 resections may include residual disease, the true recurrence risk is best assessed based on R0 cases.

Other significant factors associated with recurrence included female sex, piecemeal resection, the treatment era (from 2015 to 2020), and adenocarcinoma. Although the underlying mechanism by which female sex increases recurrence risk remains unclear, this association has been reported previously.[12] En bloc resection was associated with a lower recurrence risk, while piecemeal resection was linked to a higher risk, consistent with previous findings.[21,22] Recurrence rates were greater in the later period (2015-2020), despite similar rates of piecemeal resection, lesion size, and FAP across periods. One possible explanation is the increased proportion of procedures performed by less experienced operators (those with <20 previous EPs), which rose from 26.4% (92/349) to 44.0% (188/427). Although operator experience was not independently significant in our multivariable analysis, the binary threshold of 20 procedures likely oversimplifies the complex learning curve and may not fully capture true procedural competence. Therefore, we hypothesize that the influx of endoscopists early in their learning curve contributed to the observed increase in recurrence rates. Adenocarcinoma has been reported as a risk factor for recurrence.[23] However, some studies have reported relatively low recurrence rates in adenocarcinoma cases, suggesting that prognosis may vary based on tumor stage.[12,24] A limitation of our study is that we did not differentiate between T1a and T1b adenocarcinomas. Further investigations with more detailed staging are needed to address this. Even among patients who achieved R0 resection, the 3-year cumulative recurrence rate in our study was 11%. A previous study reported a recurrence rate of 16% (82/512) after R0 resection for ampullary tumors at a median follow-up of 20 months,[25] whereas recurrence after R0 resection of other GI epithelial tumors has been reported at 5.8% (31/532) with a mean follow-up of 13.3 months.[26] This discrepancy may be explained by the anatomical complexity of the ampullary region, which includes the sphincter of Oddi, bile duct, and pancreatic duct. These structures make accurate pathologic margin assessment more difficult.[27]

In the R0 subgroup, female sex, APC use, bile-duct invasion, and adenocarcinoma were independently associated

TABLE 4. Adverse events related to endoscopic papillectomy

	Missing	Values (n = 798)
All adverse events	0 (0%)	259 (32.5%)
Severe adverse events	0 (0%)	31 (3.9%)
Pancreatitis	0 (0%)	110 (13.8%)
Mild		63 (7.9%)
Moderate		33 (4.1%)
Severe		14 (1.8%)
Cholangitis	0 (0%)	16 (2.0%)
Mild		7 (0.9%)
Moderate		9 (1.1%)
Severe		0 (0.0%)
Bleeding	0 (0%)	155 (19.4%)
Mild		125 (15.7%)
Moderate		20 (2.5%)
Severe		10 (1.3%)
Intraprocedural perforation	0 (0%)	17 (2.1%)
Mild		7 (0.9%)
Moderate		3 (0.4%)
Severe		7 (0.9%)
Delayed perforation	0 (0%)	9 (1.1%)
Mild		3 (0.4%)
Moderate		2 (0.3%)
Severe		4 (0.5%)
Delayed biliary stenosis	0 (0%)	19 (2.4%)
Delayed pancreatic stenosis	0 (0%)	20 (2.5%)
Treatment-related mortality	0 (0%)	1 (0.1%)

Values are n (%).

with increased recurrence. APC typically is applied when there is endoscopic suspicion of residual tumor after resection. APC is typically used when there is endoscopic suspicion of residual tumor after resection. However, the reliability of endoscopic appearance during the index procedure to detect residual disease can be questionable. APC is administered when residual disease is suspected endoscopically. Notably, 17% (69/396) of patients classified as R0 received APC immediately after resection because endoscopic findings suggested possible residual disease, despite negative histologic margins on final evaluation.[25] Therefore, APC treatment itself is not considered a risk factor; rather, the presence of endoscopically suspected residual disease may contribute to the risk of recurrence. Similarly, pathologic bile duct invasion may indicate the presence of microscopic residual lesions, particularly when resection margins are limited or not clearly assessable. Post-EP pathologic invasion to bile duct similarly might indicate the possibility of remnant lesion. Similarly, adenocarcinoma remained an independent risk factor for recurrence in our analysis. Pathologic bile duct invasion also was associated with recurrence. Bile duct invasion often causes bile duct dilatation, which has been identified as a risk factor for incomplete resection[28,29] and subsequent recurrence.[13] In contrast, pancreatic duct invasion was not significant in our analysis. This may be explained by the fact that one comparison group included no recurrence events, resulting in an odds ratio with an infinite upper bound and wide CI, making the estimate statistically unstable. Importantly, this finding is in line with previous reports, which also demonstrated bile duct invasion, but not pancreatic duct invasion, as a risk factor for recurrence.[28]

Notably, neither APC use nor bile duct invasion was significantly associated with recurrence in the overall cohort, supporting the interpretation that these factors primarily reflect residual disease risk in specific subgroups, such as R0 resection cases, rather than being universal risk factors. In Rx or R1 resections, pathologic residual lesions likely have a stronger influence on recurrence than other factors. However, in the R0 resection subgroup, bile duct invasion was significantly associated with recurrence, suggesting that histologic margin negativity alone may not fully rule out residual disease when ductal involvement is present. These findings indicate that the current definition of R0—on the basis solely of negative horizontal and vertical margins—may be inadequate in cases with bile duct involvement. Incorporating the absence of pathologic ductal invasion into the definition of R0 may enhance its prognostic accuracy.

Despite the strengths of this large multicenter cohort, several limitations must be acknowledged. First, the retrospective design and 10-year study period across 22 institutions may introduce missing data and selection bias. Second, treatment strategies—including EP techniques, surveillance intervals, and APC use—varied by institution and over time, as standardized guidelines were not consistently available throughout the study. To address this, we stratified the analysis by treatment period. Third, as our study included only high-volume centers with expert teams, the generalizability of our findings may be limited. Fourth, although R0 margin status was defined using consistent pathologic criteria, evaluations were performed locally at each institution rather than centrally, which may introduce variability. Nevertheless, the recurrence rates after R0 resection were comparable to those reported in other large studies, suggesting that interobserver variability had a limited impact.

This large, multicenter cohort study shows that recurrence after EP for ampullary epithelial tumors is not uncommon, even following R0 resection. The overall 3-year cumulative recurrence rate of 20%, and 12% in R0 cases, highlights the need for ongoing postprocedural surveillance, even in patients with favorable pathologic outcomes. Given the limitations of current pathologic margin definitions—particularly in cases with ductal involvement—our findings suggest a need to refine the criteria for R0 resection in the context of ampullary

tumors. These insights support a more individualized approach to post-EP monitoring. Further prospective studies with standardized criteria are needed to validate our findings and establish uniform follow-up protocols.

DISCLOSURE

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Abbreviations: APC, argon plasma coagulation; EP, endoscopic papillectomy; FAP, familial adenomatous polyposis.

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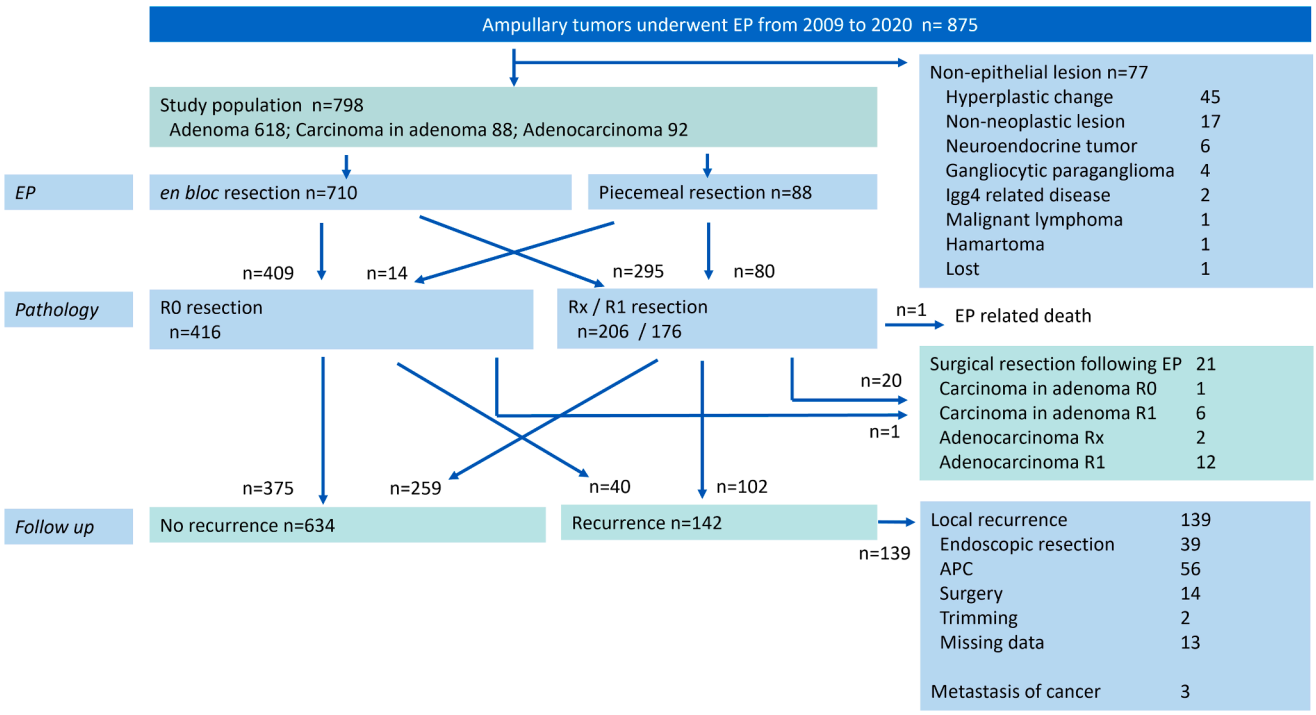
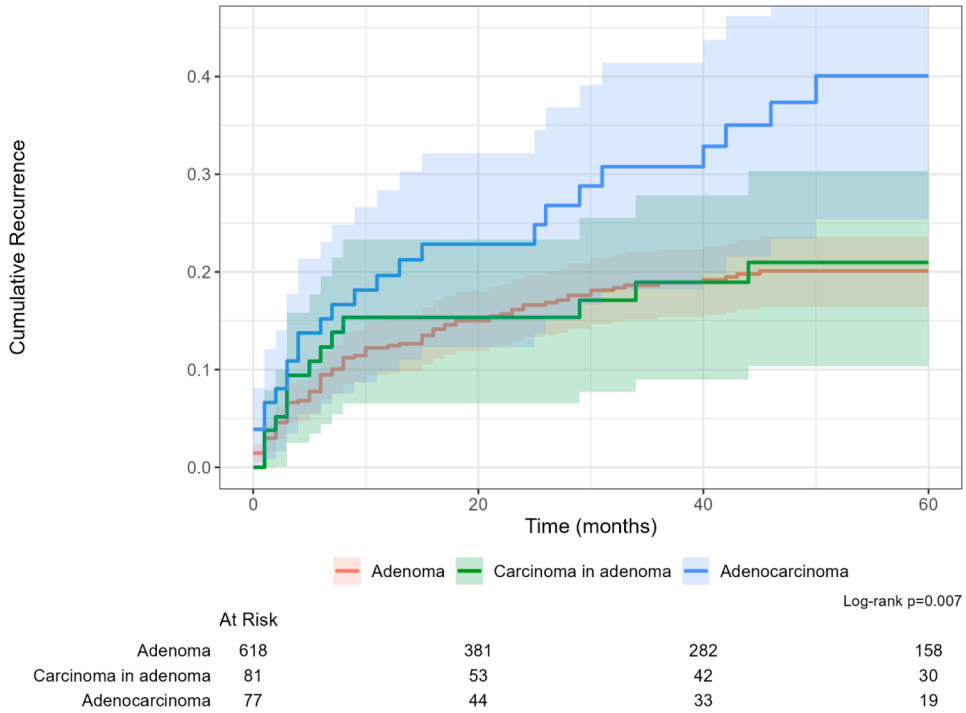
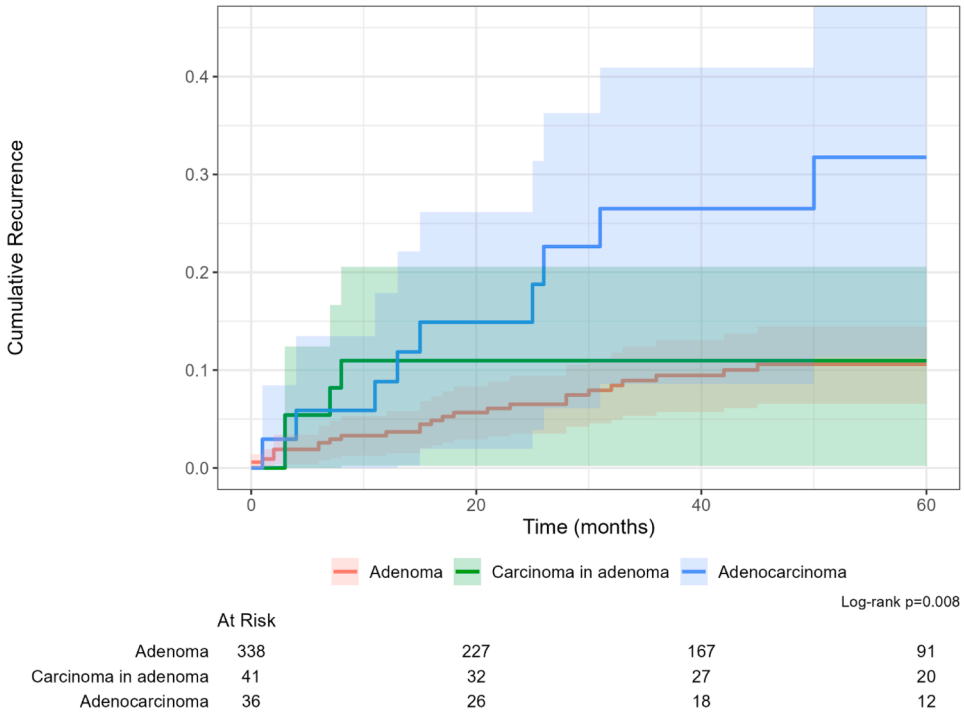


Supplementary Figure 1. The flowchart summarizing the number of patients who underwent surgery during the study period. *EP*, Endoscopic papillectomy.



Supplementary Figure 2. Kaplan–Meier curves for cumulative recurrencel in the overall cohort, stratified by adenoma, carcinoma in adenoma, and adenocarcinoma.



Supplementary Figure 3. Kaplan–Meier curves for cumulative recurrence in the R0 resection subgroup, stratified by adenoma, carcinoma in adenoma, and adenocarcinoma.

SUPPLEMENTARY TABLE 1. Additional details related to procedure

	Missing	*N* = 798
Preprocedure examination	0 (0%)	
EUS		650 (81.5%)
ERCP		465 (58.3%)
CT		557 (69.8%)
MRI		218 (27.3%)
Prophylactic pancreatic duct drainage	0 (0%)	
None		51 (6.4%)
Nasopancreatic drainage tube		13 (1.6%)
Plastic stent with internal flap		689 (86.3%)
Plastic stent without internal flap		45 (5.6%)
Diameter of plastic pancreatic stent	54 (6.8%)	
5F		691 (92.9%)
6F		1 (0.1%)
7F		52 (7.0%)
Length of plastic pancreatic stent	70 (8.8%)	
2 cm		35 (4.8%)
3 cm		204 (28.0%)
4 cm		17 (2.3%)
5 cm		363 (49.9%)
6 cm		2 (0.3%)
7 cm		107 (14.7%)
Prophylactic bile duct drainage	0 (0%)	
None		139 (17.4%)
Naso-biliary drainage tube		12 (1.5%)
Plastic stent		642 (80.5%)
Covered self-expandable metallic stent		5 (0.6%)
Diameter of plastic bile duct stent	151 (19%)	
5F		2 (0.3%)
6F		1 (0.2%)
7F		631 (97.5%)
8.5F		8 (1.2%)
10F		5 (0.8%)

Values are *n* (%).
CT, Computed tomography; *MRI*, magnetic resonance imaging.

SUPPLEMENTARY TABLE 2. Difference between 2 periods

	First period: 2009-2014	Second period: 2015-2020	
	***N* = 349**	***N* = 427**	***P* value**
Elderly	194 (55.6%)	224 (52.5%)	.4
Female	133 (38.1%)	176 (41.2%)	.4
FAP	20 (5.7%)	33 (7.7%)	.3
Large tumor (≥15 mm)	153 (43.8%)	171 (40.0%)	.3
LST-p	49 (14.0%)	72 (16.9%)	.3
Piecemeal resection	32 (9.2%)	53 (12.4%)	.2
Pathologic resection status			.004
R0	194 (55.6%)	221 (51.8%)	
R1	82 (23.5%)	75 (17.6%)	
RX	73 (20.9%)	131 (30.7%)	
Expert procedure	257 (73.6%)	239 (56.0%)	<.001
Submucosal injection	57 (16.3%)	83 (19.4%)	.3
Pathologic bile duct invasion			>.9
Positive	12 (3.4%)	14 (3.3%)	
Nonpositive	337 (96.6%)	413 (96.7%)	
Pathologic pancreatic duct invasion			.3
Positive	6 (1.7%)	3 (0.7%)	
Nonpositive	343 (98.3%)	424 (99.3%)	
Final pathologic diagnosis			.2
Adenoma	268 (76.8%)	350 (82.0%)	
Carcinoma in adenoma	41 (11.7%)	40 (9.4%)	
Adenocarcinoma	40 (11.5%)	37 (8.7%)	
Additional coagulation using APC	16 (4.6%)	8 (1.9%)	.03

Values are *n* (%). *P* values were calculated by χ^2 test or Fisher exact test, as appropriate.
APC, argon plasma coagulation; *FAP*, familial adenomatous polyposis; *LST-p*, laterally spreading tumor involving the papilla.

SUPPLEMENTARY TABLE 3. Recurrence events stratified by covariates

	No recurrence	Recurrence
	***N* = 638**	***N* = 138**
Elderly	340 (53.3%)	78 (56.5%)
Female	240 (37.6%)	69 (50.0%)
FAP	37 (5.8%)	16 (11.6%)
Large tumor (≥15 mm)	260 (40.8%)	64 (46.4%)
LST-p	87 (13.6%)	34 (24.6%)
Piecemeal resection	49 (7.7%)	36 (26.1%)
Pathologic resection status		
R0	376 (58.9%)	39 (28.3%)
R1	110 (17.2%)	47 (34.1%)
RX	152 (23.8%)	52 (37.7%)
Expert procedure	405 (63.5%)	91 (65.9%)
Submucosal injection	111 (17.4%)	29 (21.0%)
Procedure period (2015-2020)	340 (53.3%)	87 (63.0%)
Pathologic bile duct invasion		
Negative	547 (85.7%)	99 (71.7%)
Not negative	91 (14.3%)	39 (28.3%)
Pathologic pancreatic duct invasion		
Negative	559 (87.6%)	104 (75.4%)
Not negative	79 (12.4%)	34 (24.6%)
Final pathologic diagnosis		
Adenoma	518 (81.2%)	100 (72.5%)
Carcinoma in adenoma	67 (10.5%)	14 (10.1%)
Adenocarcinoma	53 (8.3%)	24 (17.4%)
Additional coagulation using APC	16 (2.5%)	8 (5.8%)

Values are *n* (%).
APC, Argon plasma coagulation; *FAP*, familial adenomatous polyposis; *LST-p*, laterally spreading tumor involving the papilla.